IPv4, IPv6 and Subnetting Practical — Answers

1. Key differences between IPv4 and IPv6

IPv4	IPv6
Uses 32-bit addresses.	Uses 128-bit addresses.
Written in dotted-decimal form, e.g., 192.168.1.10.	Written in hexadecimal groups separated by colons, e.g., 2001:db8::10.
Provides about 4.3 billion total addresses.	Provides an extremely large address space (2^128 addresses).
Subnet masks can be written as 255.255.255.0 or /24.	Uses prefix length, e.g., /64; no dotted-decimal subnet mask.
NAT is commonly used to share a public IPv4 address among devices.	Usually designed to support globally unique addressing; NAT is generally not required for address conservation.
Broadcast addressing is supported.	No broadcast; multicast and anycast are used instead.
ARP is used to discover local IPv4-to-MAC mappings.	Neighbor Discovery Protocol (NDP), using ICMPv6, performs similar local-link discovery functions.

2. Static IPv4 addresses for four home Wi-Fi devices

Network: 192.168.10.0/24. Subnet mask: 255.255.255.0. Example gateway: 192.168.10.1 (assumed to be the router's LAN address). These addresses are examples; check the router's DHCP range and reservations before applying static addresses to avoid conflicts.

Device	IP address	Subnet mask	Default gateway
Laptop	192.168.10.10	255.255.255.0	192.168.10.1
Phone	192.168.10.11	255.255.255.0	192.168.10.1
Smart TV	192.168.10.12	255.255.255.0	192.168.10.1
Printer	192.168.10.13	255.255.255.0	192.168.10.1

Valid host range for this /24: 192.168.10.1–192.168.10.254. Network address: 192.168.10.0. Broadcast address: 192.168.10.255.

3. /27 subnetting on the 192.168.1.0 network

A /27 prefix has subnet mask 255.255.255.224. It leaves 5 host bits, so each subnet contains 2^5 = 32 total addresses and 32 – 2 = 30 usable host addresses. In a /24 network divided into /27 subnets, the number of subnets is 2^(27 – 24) = 8.

Subnet	Network → Broadcast	Usable host range	Usable hosts
1	192.168.1.0 → 192.168.1.31	192.168.1.1 – 192.168.1.30	30
2	192.168.1.32 → 192.168.1.63	192.168.1.33 – 192.168.1.62	30
3	192.168.1.64 → 192.168.1.95	192.168.1.65 – 192.168.1.94	30

Assumption: “192.168.1.0 network” means the parent network 192.168.1.0/24 is being divided into /27 subnets. The first three subnets are listed above.

4. Address 10.0.5.17 with subnet mask 255.255.255.0

Step 1: Convert the subnet mask to CIDR. 255.255.255.0 = /24.

Step 2: With /24, the first three octets identify the network and the last octet identifies the host.

• Network address: 10.0.5.0

• Broadcast address: 10.0.5.255

• Valid host range: 10.0.5.1 – 10.0.5.254

• Number of usable hosts: 2^8 – 2 = 254

5. Verify with an online subnet calculator

Open https://www.subnet-calculator.com/ and enter IP address 10.0.5.17 with subnet mask 255.255.255.0 (or prefix /24). Check the displayed network address, broadcast address, and usable host range.

Suggested one-line report: “The subnet calculator showed network 10.0.5.0, broadcast 10.0.5.255, and usable hosts 10.0.5.1–10.0.5.254, so my manual answer matched the tool.”

Screenshot note: Take a screenshot of your own calculator result and insert it into your submission. This PDF does not contain a fabricated screenshot or claim that the online tool was actually run.